

Implementation Screen snapshot

Min cost, min time implementation based on your requirements

1, Storage

Amazon S3

>

Buckets

>

hc-pipeline-demo

hc-pipeline-demo

Info

Objects

Metadata

Properties

Permissions

Objects (5)

Objects are the fundamental entities stored in Amazon S3. You can use [A](#)

Find objects by prefix

☐	Name	Type
☐	 athena-results/	Folder
☐	 bronze/	Folder
☐	 gold/	Folder
☐	 raw/	Folder
☐	 silver/	Folder

Amazon S3

>

Buckets

>

hc-pipeline-demo

>

raw/

raw/

Objects

Properties

Objects (1)

Copy S3 URI

Copy URL

Download

Open

Delete

Action

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant

Find objects by prefix

☐	Name	Type	Last modified	Size
☐	 DETask.xlsx	xlsx	May 1, 2026, 20:14:53 (UTC-05:00)	664.4 KB

bronze/

Objects Properties

Objects (1)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant

☐	Find objects by prefix				
☐	Name	Type	Last modified	Size	
☐	 claims_bronze_20260502_031803.parquet	parquet	May 1, 2026, 22:18:04 (UTC-05:00)	172.8 KB	

silver/

Objects Properties

Objects (1)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant

☐	Find objects by prefix				
☐	Name	Type	Last modified	Size	
☐	 claims_silver_20260502_172631.parquet	parquet	May 2, 2026, 12:26:33 (UTC-05:00)	214.6 KB	

gold/

Objects Properties

Objects (2)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#)

☐	Find objects by prefix				
☐	Name	Type			
☐	 iceberg/	Folder			
☐	 staging/	Folder			

staging/

Objects Properties

Objects (1)

Copy S3 URI

Copy URL

Download

Open ↗

Delete

Action

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant

Find objects by prefix

☐	Name	Type	Last modified	Size
☐	data_20260502_181431.parquet	parquet	May 2, 2026, 13:14:34 (UTC-05:00)	214.6 KB

[Amazon S3](#) > [Buckets](#) > [hc-pipeline-demo](#) > [gold/](#) > [iceberg/](#)

iceberg/

Objects

Properties

Objects (7)

Objects are the fundamental entities stored in Amazon S3. You can use [An](#)

🔍 Find objects by prefix

☐	Name	▲	Type
☐	📁 dim_date/		Folder
☐	📁 dim_member/		Folder
☐	📁 dim_payer/		Folder
☐	📁 dim_procedure/		Folder
☐	📁 dim_provider/		Folder
☐	📁 fact_claims/		Folder
☐	📁 fact_remittance/		Folder

fact_claims/

Objects | Properties

Objects (2)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3](#)

Find objects by prefix

☐	Name	Type
☐	 data/	Folder
☐	 metadata/	Folder

metadata/

Objects | Properties

Objects (4)

Copy S3 URICopy URLDownloadOpen iDeleteActions

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory i](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant t

Find objects by prefix

☐	Name	Type	Last modified	Size
☐	 00000-722c95be-699a-4ffd-9896-1f6cb0d525e9.metadata.json	json	May 2, 2026, 13:14:39 (UTC-05:00)	2.5 KB
☐	 00001-3cd8934d-c28c-4ec9-8ed7-3a817c64738b.metadata.json	json	May 2, 2026, 13:14:48 (UTC-05:00)	3.5 KB
☐	 b33b6415-b31e-4426-93ac-509429606df9-m0.avro	avro	May 2, 2026, 13:14:47 (UTC-05:00)	9.7 KB
☐	 snap-2662624071159128714-1-b33b6415-b31e-4426-93ac-509429606df9.avro	avro	May 2, 2026, 13:14:47 (UTC-05:00)	4.2 KB

data/

Objects Properties

Objects (12)   Copy

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permission.

Find objects by prefix

☐	Name	Type
☐	_C5kQ/	Folder
☐	0BouwW/	Folder
☐	73hoUQ/	Folder
☐	9C_YNw/	Folder
☐	d7Sf5A/	Folder
☐	jy2FrA/	Folder
☐	nk0eww/	Folder
☐	O5qmhA/	Folder
☐	RVXs6g/	Folder
☐	t8NiKw/	Folder
☐	wUnUJw/	Folder
☐	YLVwFg/	Folder

month=10/

Objects Properties

Objects (1)   Copy S3 URI  Copy URL  Download  Open  Delete  Action

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permission.

Find objects by prefix

☐	Name	Type	Last modified	Size
☐	20260502_181444_00088_mtv83-74f34718-f87c-478f-af2b-d26e480bb356.parquet	parquet	May 2, 2026, 13:14:47 (UTC-05:00)	11.9 KB

2, Lambda computing

≡ [Lambda](#) > Functions

Functions (3) Last fetched 5/2/2026, 4:11:44 PM [Actions](#) [Cr](#)

Q Search by attributes or search by keyword

☐	Function name	Description	Package type	Runtime	Type	Last modified
☐	gold-transformation	-	Zip	Python 3.12	Standard	3 hours ago
☐	bronze-ingestion	-	Zip	Python 3.12	Standard	19 hours ago
☐	silver-clean	-	Zip	Python 3.12	Standard	4 hours ago

Use built-in layer

Layers (1) [Info](#) Last fetched 5/2/2026, 4:12:16 PM [1](#)

Q Search by attributes or search by keyword

Merge order	Name	Layer version	Compatible runtimes	Compatible architectures	Version ARN
1	AWSSDKPandas-Python312	24	python3.12	x86_64	arn:aws:lambda:us-east-1:336392948345:layer:AWSSDKPandas-Python312:24

Use different role, gold-lambda use gold-lambda-role; brone and silver use same role called “[bronze-ingestion-role-b82ljmie](#)”

≡ [Lambda](#) > [Functions](#) > gold-transformation

[Code](#) [Test](#) [Monitor](#) [Configuration](#) [Aliases](#) [Versions](#)

Configuration

- General configuration
- Triggers
- Permissions**
- Destinations
- Function URL
- Environment variables
- Tags
- VPC
- RDS databases
- Monitoring and operations tools
- Concurrency and recursion detection
- Asynchronous invocation

Execution role [View role document](#)

Role name
[gold-lambda-role](#)

Resource summary
To view the resources and actions that your function has permission to access, choose a service.

[Amazon Athena](#)
4 actions 1 resource

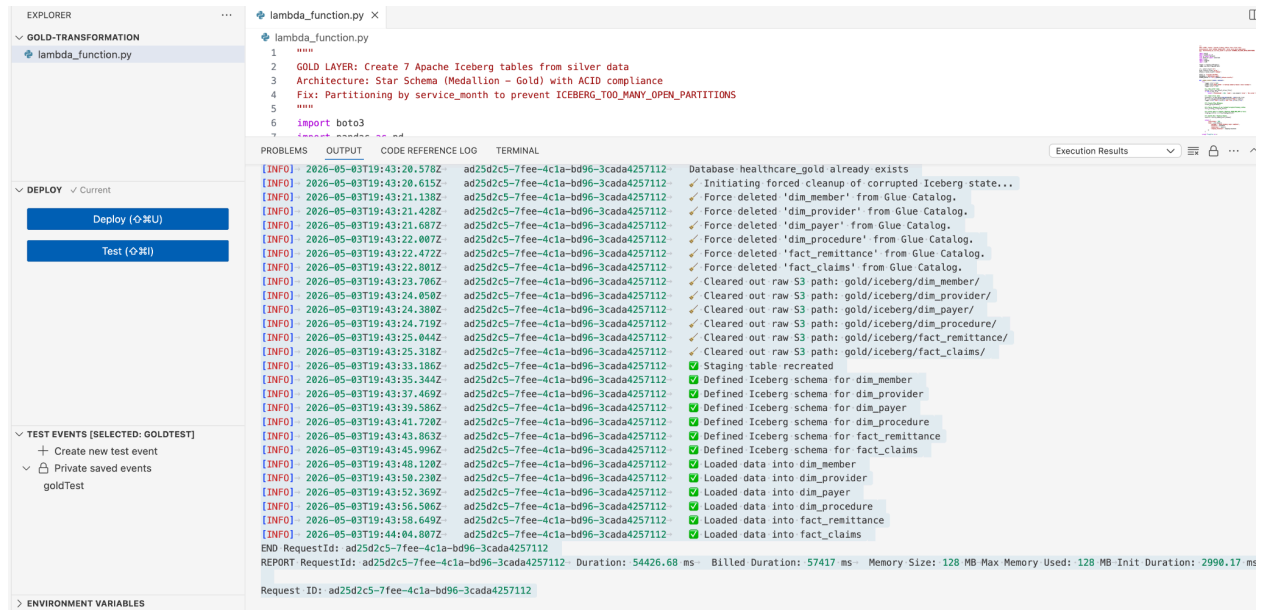
By action **By resource**

Resource	Actions
All resources	Allow: athena:StartQueryExecution Allow: athena:GetQueryExecution Allow: athena:GetQueryResults Allow: athena:StopQueryExecution

Lambda obtained this information from the following policy statements:

- Managed policy gold-lambda-policy, statement 2

Real time monitoring when running gold lambda



Response:

```
{
  "statusCode": 200,
  "body": "{\"message\": \"Gold Iceberg layer complete\", \"database\": \"healthcare_gold\", \"executions\": {\"dim_member\": \"SUCCESS\", \"dim_provider\": \"SUCCESS\", \"dim_payer\": \"SUCCESS\", \"dim_procedure\": \"SUCCESS\", \"fact_remittance\": \"SUCCESS\", \"fact_claims\": \"SUCCESS\", \"dim_date\": \"ERROR: Athena query FAILED: HIVE_PATH_ALREADY_EXISTS: Target directory for table 'healthcare_gold.dim_date' already exists: s3://hc-pipeline-demo/gold/iceberg/dim_date/. You may need to manually clean the data at location 's3://hc-pipeline-demo/athena-results/tables/claa5986-3536-4786-8ecf-fc294ef6bee8' before retrying. Athena will not delete data in your account.\"}, \"staging_location\": \"s3://hc-pipeline-demo/gold/staging/data_20260503_194325.parquet\"}"
}
```

The area below shows the last 4 KB of the execution log.

Function Logs:

```
[INFO] 2026-05-03T19:43:14.480Z Found credentials in environment variables.
START RequestId: ad25d2c5-7fee-4c1a-bd96-3cada4257112 Version: $LATEST
[INFO] 2026-05-03T19:43:14.730Z ad25d2c5-7fee-4c1a-bd96-3cada4257112
=====
[INFO] 2026-05-03T19:43:14.730Z ad25d2c5-7fee-4c1a-bd96-3cada4257112 GOLD LAYER
- Creating Iceberg Tables (Star Schema)
```



```

[INFO] 2026-05-03T19:43:14.730Z      ad25d2c5-7fee-4c1a-bd96-3cada4257112
=====
[INFO] 2026-05-03T19:43:20.097Z      ad25d2c5-7fee-4c1a-bd96-3cada4257112      Read 8381
rows from silver/claims_silver_20260502_172631.parquet
[INFO] 2026-05-03T19:43:20.578Z      ad25d2c5-7fee-4c1a-bd96-3cada4257112      Database
healthcare_gold already exists
[INFO] 2026-05-03T19:43:20.615Z      ad25d2c5-7fee-4c1a-bd96-3cada4257112      🛠️
Initiating forced cleanup of corrupted Iceberg state...
[INFO] 2026-05-03T19:43:21.138Z      ad25d2c5-7fee-4c1a-bd96-3cada4257112      🛠️ Force
deleted 'dim_member' from Glue Catalog.
[INFO] 2026-05-03T19:43:21.428Z      ad25d2c5-7fee-4c1a-bd96-3cada4257112      🛠️ Force
deleted 'dim_provider' from Glue Catalog.
[INFO] 2026-05-03T19:43:21.687Z      ad25d2c5-7fee-4c1a-bd96-3cada4257112      🛠️ Force
deleted 'dim_payer' from Glue Catalog.
[INFO] 2026-05-03T19:43:22.007Z      ad25d2c5-7fee-4c1a-bd96-3cada4257112      🛠️ Force
deleted 'dim_procedure' from Glue Catalog.
[INFO] 2026-05-03T19:43:22.472Z      ad25d2c5-7fee-4c1a-bd96-3cada4257112      🛠️ Force
deleted 'fact_remittance' from Glue Catalog.
[INFO] 2026-05-03T19:43:22.801Z      ad25d2c5-7fee-4c1a-bd96-3cada4257112      🛠️ Force
deleted 'fact_claims' from Glue Catalog.
[INFO] 2026-05-03T19:43:23.706Z      ad25d2c5-7fee-4c1a-bd96-3cada4257112      🛠️ Cleared
out raw S3 path: gold/iceberg/dim_member/
[INFO] 2026-05-03T19:43:24.050Z      ad25d2c5-7fee-4c1a-bd96-3cada4257112      🛠️ Cleared
out raw S3 path: gold/iceberg/dim_provider/
[INFO] 2026-05-03T19:43:24.380Z      ad25d2c5-7fee-4c1a-bd96-3cada4257112      🛠️ Cleared
out raw S3 path: gold/iceberg/dim_payer/
[INFO] 2026-05-03T19:43:24.719Z      ad25d2c5-7fee-4c1a-bd96-3cada4257112      🛠️ Cleared
out raw S3 path: gold/iceberg/dim_procedure/
[INFO] 2026-05-03T19:43:25.044Z      ad25d2c5-7fee-4c1a-bd96-3cada4257112      🛠️ Cleared
out raw S3 path: gold/iceberg/fact_remittance/
[INFO] 2026-05-03T19:43:25.318Z      ad25d2c5-7fee-4c1a-bd96-3cada4257112      🛠️ Cleared
out raw S3 path: gold/iceberg/fact_claims/
[INFO] 2026-05-03T19:43:33.186Z      ad25d2c5-7fee-4c1a-bd96-3cada4257112      ✅ Staging
table recreated
[INFO] 2026-05-03T19:43:35.344Z      ad25d2c5-7fee-4c1a-bd96-3cada4257112      ✅ Defined
Iceberg schema for dim_member
[INFO] 2026-05-03T19:43:37.469Z      ad25d2c5-7fee-4c1a-bd96-3cada4257112      ✅ Defined
Iceberg schema for dim_provider
[INFO] 2026-05-03T19:43:39.586Z      ad25d2c5-7fee-4c1a-bd96-3cada4257112      ✅ Defined
Iceberg schema for dim_payer
[INFO] 2026-05-03T19:43:41.720Z      ad25d2c5-7fee-4c1a-bd96-3cada4257112      ✅ Defined
Iceberg schema for dim_procedure

```

```

[INFO] 2026-05-03T19:43:43.863Z ad25d2c5-7fee-4c1a-bd96-3cada4257112 ✓ Defined
Iceberg schema for fact_remittance
[INFO] 2026-05-03T19:43:45.996Z ad25d2c5-7fee-4c1a-bd96-3cada4257112 ✓ Defined
Iceberg schema for fact_claims
[INFO] 2026-05-03T19:43:48.120Z ad25d2c5-7fee-4c1a-bd96-3cada4257112 ✓ Loaded
data into dim_member
[INFO] 2026-05-03T19:43:50.230Z ad25d2c5-7fee-4c1a-bd96-3cada4257112 ✓ Loaded
data into dim_provider
[INFO] 2026-05-03T19:43:52.369Z ad25d2c5-7fee-4c1a-bd96-3cada4257112 ✓ Loaded
data into dim_payer
[INFO] 2026-05-03T19:43:56.506Z ad25d2c5-7fee-4c1a-bd96-3cada4257112 ✓ Loaded
data into dim_procedure
[INFO] 2026-05-03T19:43:58.649Z ad25d2c5-7fee-4c1a-bd96-3cada4257112 ✓ Loaded
data into fact_remittance
[INFO] 2026-05-03T19:44:04.807Z ad25d2c5-7fee-4c1a-bd96-3cada4257112 ✓ Loaded
data into fact_claims
END RequestId: ad25d2c5-7fee-4c1a-bd96-3cada4257112
REPORT RequestId: ad25d2c5-7fee-4c1a-bd96-3cada4257112 Duration: 54426.68 ms Billed
Duration: 57417 ms Memory Size: 128 MB Max Memory Used: 128 MB Init Duration:
2990.17 ms

```

Request ID: ad25d2c5-7fee-4c1a-bd96-3cada4257112

3, IAM

Identity and Access Management (IAM)

Roles (5)

An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assigned to AWS services, applications, or users.

Role name	Trusted entities	Last activity
AWSServiceRoleForResourceExplorer	AWS Service: resource-explorer-2 (Service-Linked Role)	6 hours ago
AWSServiceRoleForSupport	AWS Service: support (Service-Linked Role)	-
AWSServiceRoleForTrustedAdvisor	AWS Service: trustedadvisor (Service-Linked Role)	-
bronze-ingestion-role-b82ljmie	AWS Service: lambda	6 hours ago
gold-lambda-role	AWS Service: lambda	6 hours ago

4, Glue Data Catalog,

AWS Glue > **Databases** > healthcare_gold

healthcare_gold Last updated (UTC) May 3, 2026 at 20:26:18 [Edit](#) [Delete](#)

Database properties

Name healthcare_gold	Description Healthcare claims Gold layer - Iceberg tables	Location -	Created on (UTC) May 2, 2026 at 14:34:11
--------------------------------	---	----------------------	--

Tables (8) Last updated (UTC) May 3, 2026 at 20:26:19 [Delete](#) [Add tables using crawler](#) [Add table](#)

View and manage all available tables.

☐	Name	Database	Location	Classification	Deprecated	View data	Data quality	Column statistics
☐	dim_date	healthcare_gold	s3://hc-pipeline-demo/	-	-	Table data	View data quality	View statistics
☐	dim_member	healthcare_gold	s3://hc-pipeline-demo/	-	-	Table data	View data quality	View statistics
☐	dim_payer	healthcare_gold	s3://hc-pipeline-demo/	-	-	Table data	View data quality	View statistics
☐	dim_procedure	healthcare_gold	s3://hc-pipeline-demo/	-	-	Table data	View data quality	View statistics
☐	dim_provider	healthcare_gold	s3://hc-pipeline-demo/	-	-	Table data	View data quality	View statistics
☐	fact_claims	healthcare_gold	s3://hc-pipeline-demo/	-	-	Table data	View data quality	View statistics
☐	fact_remittance	healthcare_gold	s3://hc-pipeline-demo/	-	-	Table data	View data quality	View statistics
☐	staging_claims	healthcare_gold	s3://hc-pipeline-demo/	-	-	Table data	View data quality	View statistics

AWS Glue > **Databases**

Databases (4) Last updated (UTC) May 3, 2026 at 02:05:44 [Edit](#) [Delete](#) [Add database](#)

A database is a set of associated table definitions, organized into a logical group.

Choose catalog
899957567554

☐	Name	Description	Location URI	Source catalog	Created on (UTC)
☐	default	-	-	899957567554	May 2, 2026 at 14:51:28
☐	healthcare_gold	Healthcare claims Gold layer - Iceberg ti	-	899957567554	May 2, 2026 at 14:34:11
☐	healthcare_silver	-	-	899957567554	May 2, 2026 at 21:57:07
☐	sagemaker_sample_db	-	-	899957567554	May 2, 2026 at 22:26:20

AWS Glue supports multiple table optimization options to enhance the management and performance of Apache Iceberg tables used by the AWS analytical engines and ETL jobs. These optimizers provide efficient storage utilization, improved query performance, and effective data management; Catalog-level optimization configuration is available through the Lake Formation console and using the AWS Glue `UpdateCatalog` API operation. You can enable or disable compaction, snapshot retention, and orphan file deletion optimizers for individual Iceberg tables in the Data Catalog using the AWS Glue console, AWS CLI, or AWS Glue API operations.

5, Athena –semantic layer for tableau/QuickSight/PowerBI or other BI tools

Data

Data source

AwsDataCatalog

Catalog

None

Database

healthcare_gold

Tables and views

Create

Filter tables and views

Tables (8)

dim_date

dim_member

dim_payer

dim_procedure

dim_provider

fact_claims

fact_remittance

staging_claims

Views (0)

Query 2

```

1 SELECT 'dim_member' AS table_name, COUNT(*) AS total_rows FROM healthcare_gold.dim_member
2 UNION ALL
3 SELECT 'dim_provider' AS table_name, COUNT(*) AS total_rows FROM healthcare_gold.dim_provider
4 UNION ALL
5 SELECT 'dim_payer' AS table_name, COUNT(*) AS total_rows FROM healthcare_gold.dim_payer
6 UNION ALL
7 SELECT 'dim_procedure' AS table_name, COUNT(*) AS total_rows FROM healthcare_gold.dim_procedure
8 UNION ALL
9 SELECT 'dim_date' AS table_name, COUNT(*) AS total_rows FROM healthcare_gold.dim_date
10 UNION ALL
11 SELECT 'fact_claims' AS table_name, COUNT(*) AS total_rows FROM healthcare_gold.fact_claims
12 UNION ALL
13 SELECT 'fact_remittance' AS table_name, COUNT(*) AS total_rows FROM healthcare_gold.fact_remittance
14 ORDER BY total_rows DESC;

```

SQL Ln 14, Col 26

Run again

Explain

Cancel

Clear

Create

Query results

Query stats

Completed

Results (7)

Search rows

#	table_name	total_rows
1	fact_claims	8381
2	dim_date	4018
3	fact_remittance	3678
4	dim_member	990
5	dim_procedure	431
6	dim_provider	366
7	dim_payer	3

6, Views for a Complete Semantic Layer

Semantic Layer Usage Examples

To demonstrate the business value of the Gold layer, I have created role-specific Semantic Views. These views abstract the complexity of the Star Schema away from end-users and BI tools.

Example Business Scenario: Identifying Revenue Leakage A Billing Manager wants to know why revenue is being lost to insurance adjustments. Instead of writing complex JOINS across millions of rows, they simply query the pre-aggregated view:

```

SELECT carc_code, frequency_of_adjustment, estimated_revenue_impact
FROM healthcare_gold.vw_claim_adjustment_analysis
ORDER BY estimated_revenue_impact DESC
LIMIT 3;

```

ALL SQL were tested and run successfully

Final Result: The Semantic Layer in Action

6.1. Executive Financial Summary (Monthly Trends)

Purpose: Provides a high-level view of monthly revenue, total write-offs (discounts), and payment ratios.

```
CREATE OR REPLACE VIEW healthcare_gold.vw_executive_financial_summary AS
SELECT
    d.year,
    d.month,
    f.payer_name,
    COUNT(DISTINCT f.claim_id) AS total_claims,
    SUM(f.total_charges) AS gross_charges,
    SUM(f.allowed_amt) AS total_allowed,
    SUM(f.total_charges - f.allowed_amt) AS total_discount_amt,
    COALESCE(SUM(r.payment_amount), 0) AS total_payments_received,
    ROUND(SUM(f.allowed_amt) / NULLIF(SUM(f.total_charges), 0) * 100, 2) AS
payment_to_charge_ratio_pct
FROM healthcare_gold.fact_claims f
JOIN healthcare_gold.dim_date d ON f.service_date_key = d.date_key
LEFT JOIN (
    -- Pre-aggregate payments to avoid fan-out from multiple adjustments
    SELECT claim_id, SUM(payment_amount) as payment_amount
    FROM healthcare_gold.fact_remittance
    GROUP BY claim_id
) r ON f.claim_id = r.claim_id
GROUP BY d.year, d.month, f.payer_name
ORDER BY d.year DESC, d.month DESC;
```

To see the Monthly Revenue Trend:

Query 2

```
1 SELECT * FROM healthcare_gold.vw_executive_financial_summary
2 ORDER BY year DESC, month DESC;
```

SQL Ln 2, Col 32

Run again Explain Clear Create

Query results Query stats

Completed Time in queue: 99 ms

Results (33)

Search rows

#	year	month	payer_name	total_claims	gross_charges	total_allowed	total_discount_amt	total_payments_received	payment_to_charge_ratio_pct
1	2021	12	Group	111	24260.88	19465.50	4795.38	6663.50	80.00
2	2021	12	Medicare	49	151627.89	17104.04	134523.85	8515.00	11.00
3	2021	12		1510	553483.43	360315.99	173167.44	157414.08	68.00
4	2021	11	Medicare	32	4646.90	4270.00	378.90	2325.00	93.00
5	2021	11		1413	417920.27	258106.13	159814.14	120503.24	62.00
6	2021	11	Group	76	17281.20	14825.46	2455.74	1165.00	86.00
7	2021	10		1084	239163.83	145314.45	93851.38	70448.50	69.00
8	2021	10	Medicare	51	10761.34	8680.00	2081.34	3854.00	81.00
9	2021	10	Group	51	9750.76	8360.00	1410.76	7486.00	86.00
10	2021	9	Group	14	2385.38	1859.00	526.38	785.00	84.00
11	2021	9		1061	367181.26	226363.62	140795.64	88348.14	62.00
12	2021	9	Medicare	42	18110.89	13204.80	4806.09	5413.54	73.00
13	2021	8	Group	33	8085.00	6035.00	2050.00	1450.00	75.00
14	2021	8	Medicare	22	3980.38	3310.00	670.38	3065.00	83.00
15	2021	8		576	173239.99	138088.11	35151.88	70483.50	80.00
16	2021	7	Group	30	4176.50	3698.50	478.00	1187.50	89.00
17	2021	7		203	54822.85	41617.99	13204.86	20471.18	76.00
18	2021	7	Medicare	2	240.00	240.00	0.00	0.00	100.00
19	2021	6	Medicare	33	2709.07	2314.07	395.00	700.00	93.00
20	2021	6		788	147154.38	113748.50	33405.88	7642.00	91.00
21	2021	6	Group	11	5839.01	2972.00	2867.01	1496.00	51.00
22	2021	6	Individual	3	445.58	438.00	7.58	0.00	98.00

6.2. Provider Performance & Network Analysis

Purpose: Evaluates which providers are generating the most volume and how much of their billing is in-network vs. out-of-network.

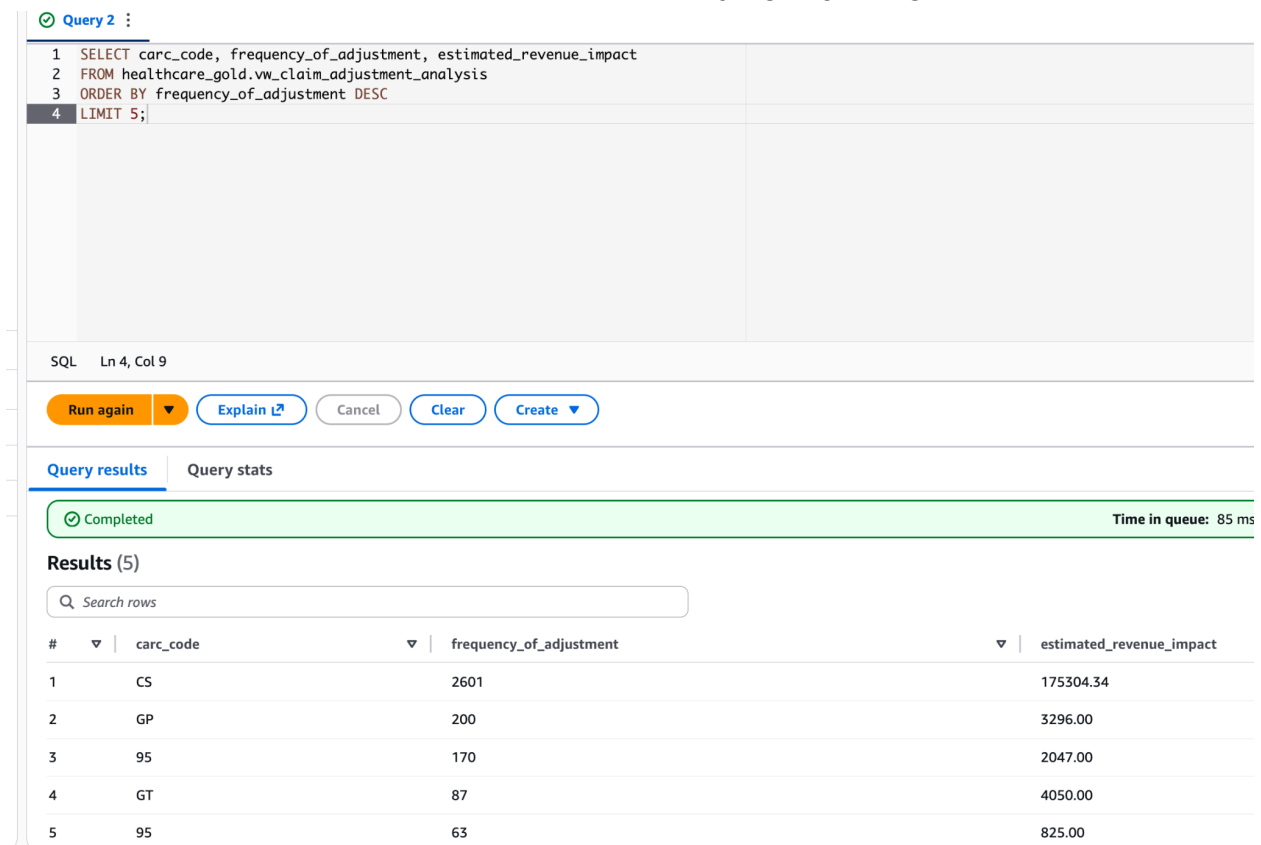
```
CREATE OR REPLACE VIEW healthcare_gold.vw_provider_performance AS
SELECT
    p.provider_name,
    p.city,
    p.state,
    f.in_network,
    COUNT(f.claim_id) AS claim_volume,
    SUM(f.units_count) AS total_services_rendered,
    SUM(f.total_charges) AS total_billed_revenue,
    ROUND(AVG(f.total_charges), 2) AS avg_charge_per_claim
FROM healthcare_gold.fact_claims f
JOIN healthcare_gold.dim_provider p ON f.provider_npi = p.provider_npi
GROUP BY p.provider_name, p.city, p.state, f.in_network
ORDER BY total_billed_revenue DESC;
```

6.3. Claim Adjustment & Denial Analysis

Purpose: Identifies the most common reasons insurance companies are reducing payments (using CARC codes).

```
CREATE OR REPLACE VIEW healthcare_gold.vw_claim_adjustment_analysis AS
SELECT
    f.payer_name,
    r.carc_code,
    COUNT(r.claim_id) AS frequency_of_adjustment,
    SUM(f.total_charges - f.allowed_amt) AS estimated_revenue_impact
FROM healthcare_gold.fact_remittance r
JOIN healthcare_gold.fact_claims f ON r.claim_id = f.claim_id
WHERE r.carc_code IS NOT NULL
GROUP BY f.payer_name, r.carc_code
ORDER BY frequency_of_adjustment DESC;
```

To find the most common reason insurance is denying/adjusting claims:



Query 2 :

```
1 SELECT carc_code, frequency_of_adjustment, estimated_revenue_impact
2 FROM healthcare_gold.vw_claim_adjustment_analysis
3 ORDER BY frequency_of_adjustment DESC
4 LIMIT 5;
```

SQL Ln 4, Col 9

Run again Explain Cancel Clear Create

Query results Query stats

Completed Time in queue: 85 ms

Results (5)

Search rows

#	carc_code	frequency_of_adjustment	estimated_revenue_impact
1	CS	2601	175304.34
2	GP	200	3296.00
3	95	170	2047.00
4	GT	87	4050.00
5	95	63	825.00

6.4, 4. Procedure Profitability Matrix

Purpose: Shows which medical procedures (CPT/HCPCS codes) are billed most frequently and yield the highest allowed reimbursement rates.

```
CREATE OR REPLACE VIEW healthcare_gold.vw_procedure_financials AS
SELECT
    dp.procedure_code,
    dp.procedure_desc,
    COUNT(f.claim_id) AS times_billed,
    SUM(f.units_count) AS total_units,
    ROUND(AVG(f.total_charges), 2) AS avg_billed_amount,
    ROUND(AVG(f.allowed_amt), 2) AS avg_allowed_amount,
    ROUND(SUM(f.allowed_amt) / NULLIF(SUM(f.total_charges), 0) * 100, 2) AS
yield_percentage
FROM healthcare_gold.fact_claims f
JOIN healthcare_gold.dim_procedure dp ON f.procedure_code = dp.procedure_code
GROUP BY dp.procedure_code, dp.procedure_desc
ORDER BY times_billed DESC;
```

To find the Top 5 Most Billed Procedures:

The screenshot shows a SQL query execution interface. The query is as follows:

```
1 SELECT procedure_code, procedure_desc, times_billed, avg_allowed_amount
2 FROM healthcare_gold.vw_procedure_financials
3 ORDER BY times_billed DESC
4 LIMIT 5;
```

Below the query editor, there are buttons for "Run again", "Explain", "Cancel", "Clear", and "Create". The "Run again" button is highlighted.

The "Query results" tab is selected, showing a status bar that says "Completed" with a green checkmark. To the right of the status bar, it says "Time in queue: 128 ms" and "Run time: 128 ms".

Below the status bar, there is a section titled "Results (5)" with a search bar. The results are displayed in a table with the following columns: #, procedure_code, procedure_desc, times_billed, and avg_allowed_amount.

#	procedure_code	procedure_desc	times_billed	avg_allowed_amount
1	U0005		1970	73.31
2	U0003		1495	182.60
3	G2023		1389	59.81
4	U0004		732	257.96
5	90834		655	170.51

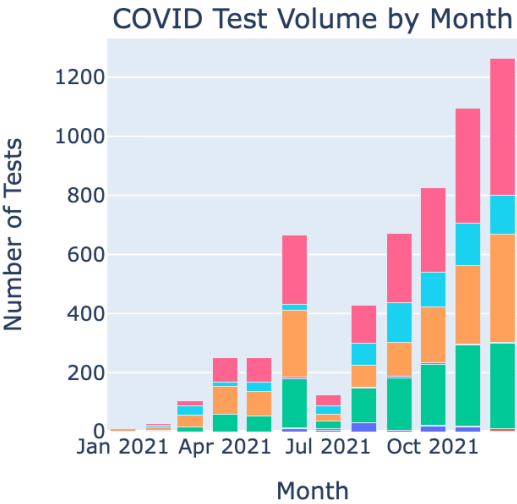
Using those views and create some Vizs as below, created by Sagemake notebook,



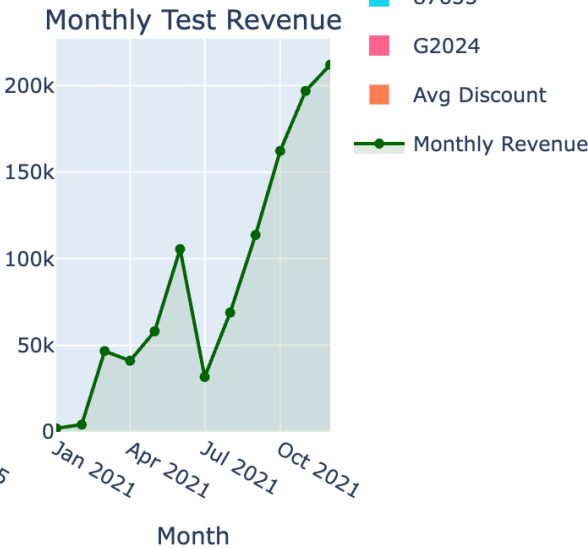
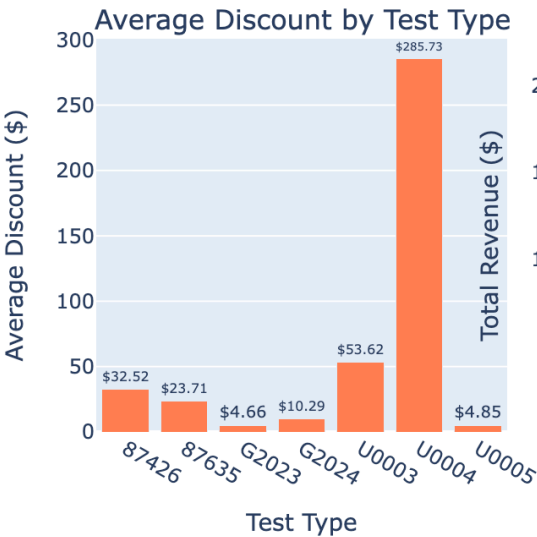
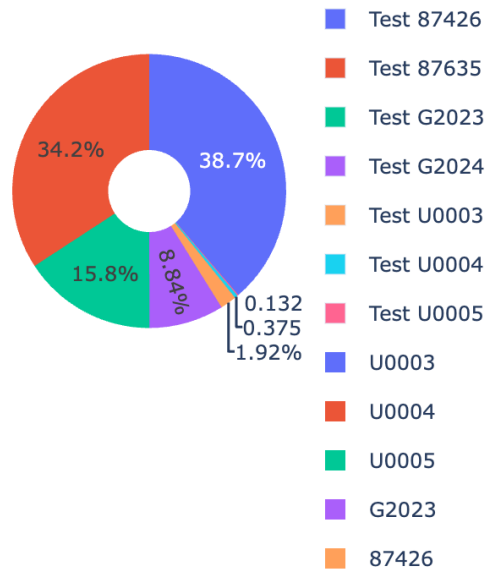
Healthcare Financial Dashboard



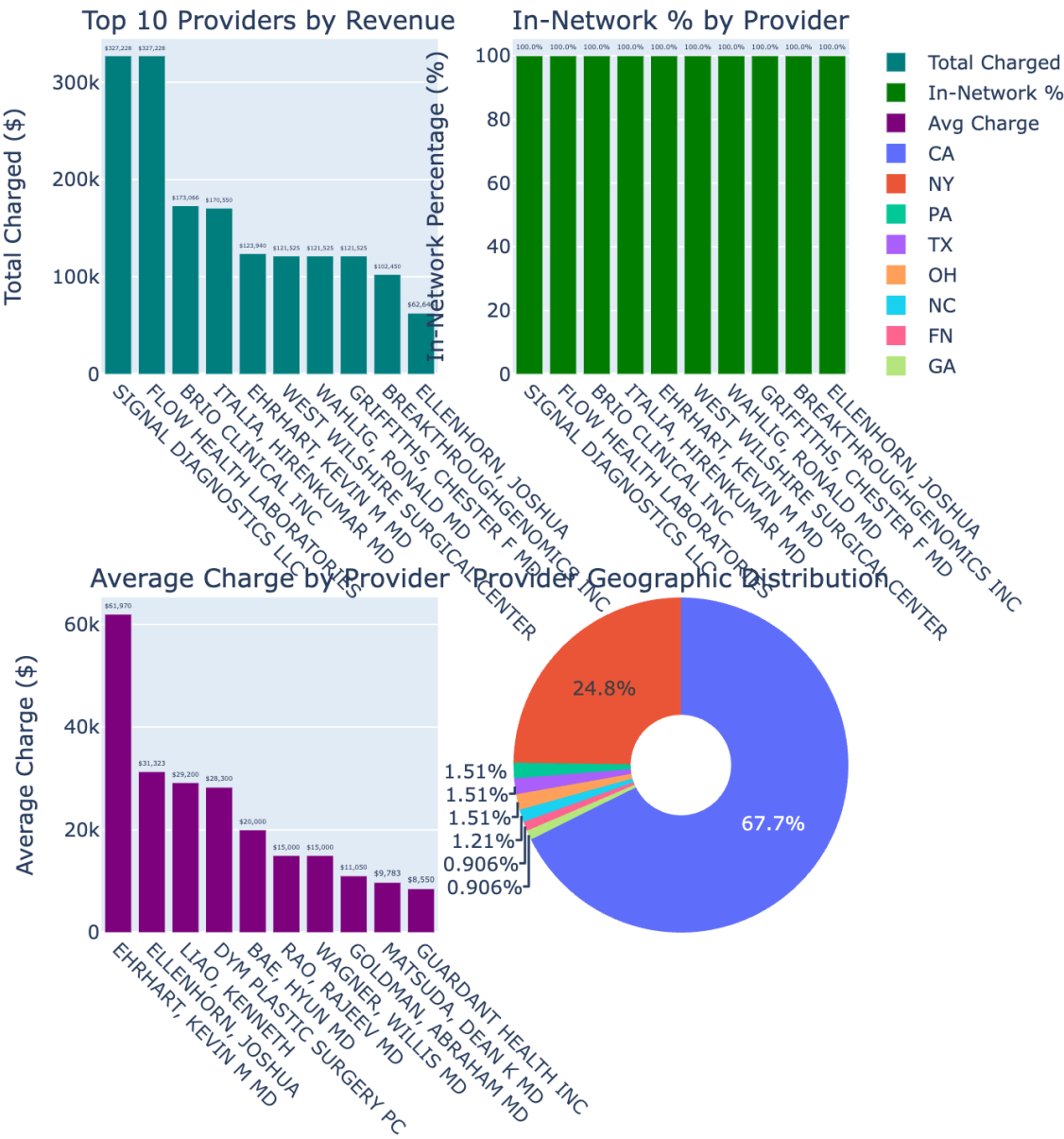
COVID-19 Testing Analytics



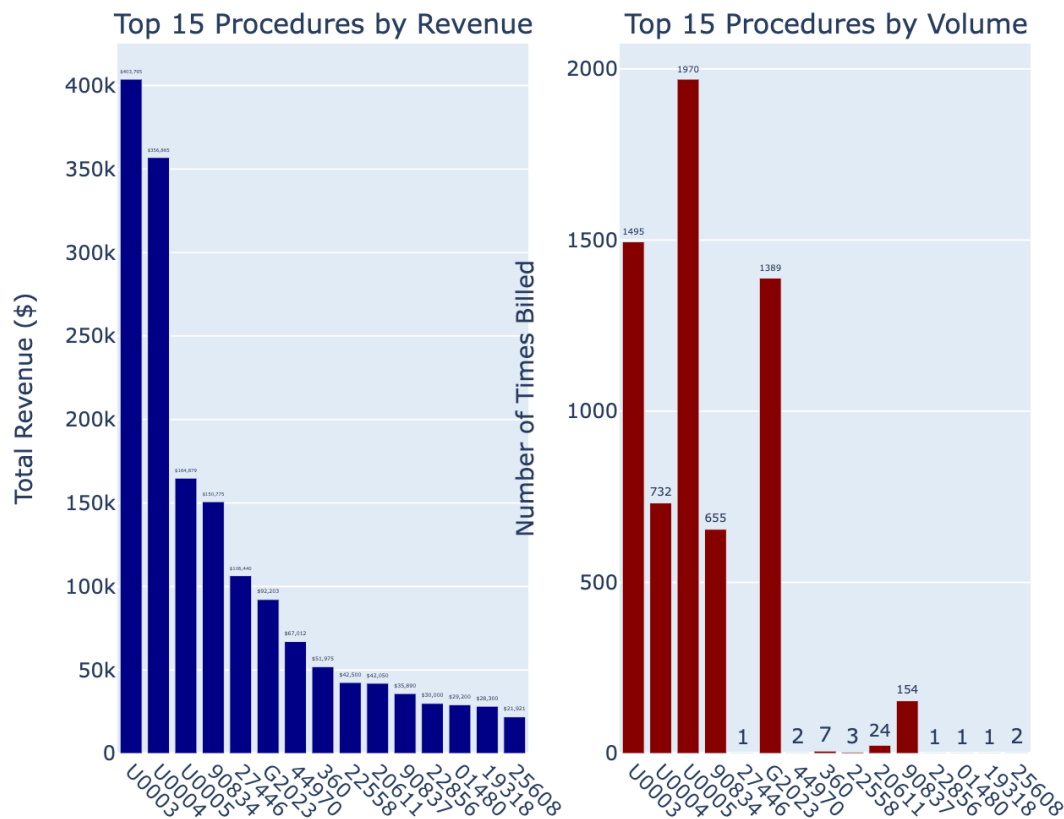
Revenue by Test Type



Provider Performance Analytics



Procedure Analysis



=====
✍ Procedure Analysis Summary
=====

Total unique procedures: 431
Total revenue across all procedures: \$2,565,170.32
Total claims across all procedures: 8,381

🏆 Top 5 Procedures by Revenue:
U0003:
Revenue: \$403,795.46
Claims: 1,495
Units: 1,495

U0004:

Revenue: \$356,885.25

Claims: 732

Units: 732

U0005:

Revenue: \$164,879.25

Claims: 1,970

Units: 1,970

90834:

Revenue: \$150,775.00

Claims: 655

Units: 665

27446:

Revenue: \$106,440.00

Claims: 1

Units: 1

💰 Highest Revenue per Claim Procedures:

27446: \$106,440.00 per claim

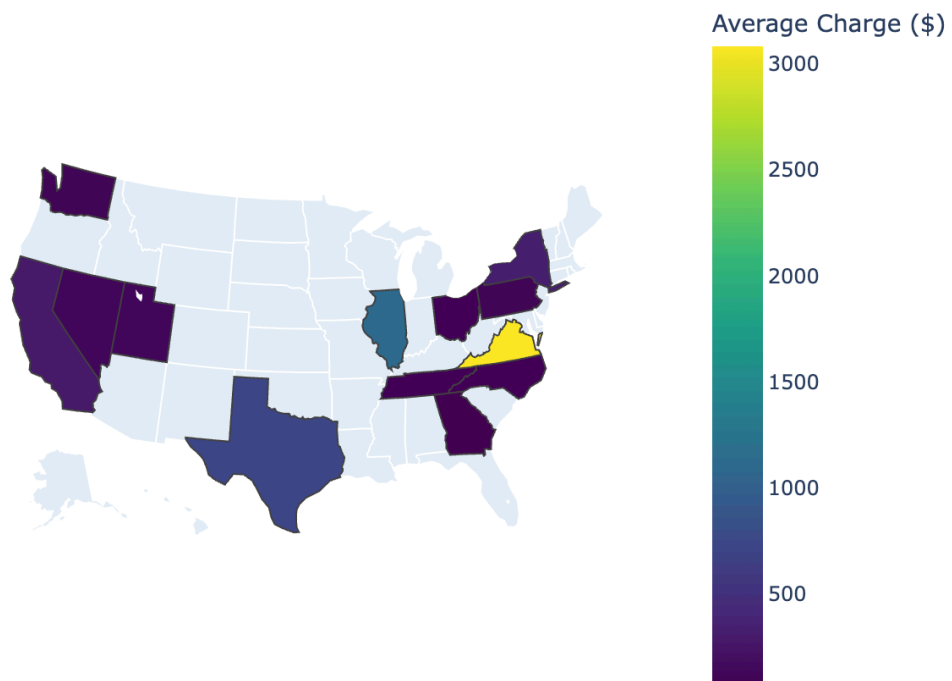
44970: \$33,505.90 per claim

22856: \$30,000.00 per claim

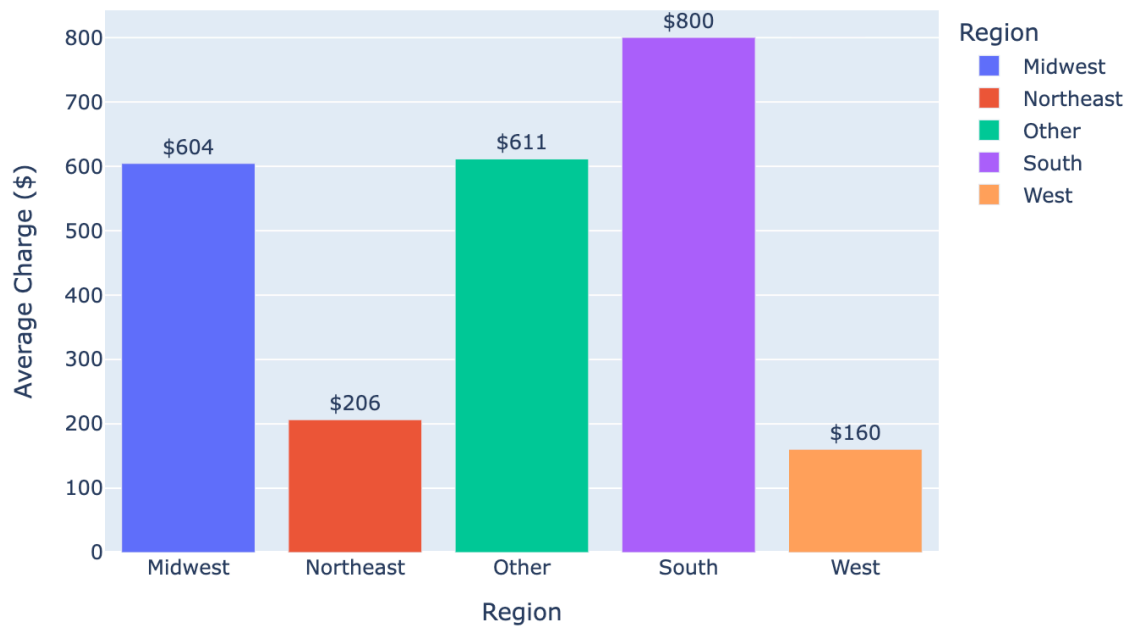
01480: \$29,200.00 per claim

19318: \$28,300.00 per claim

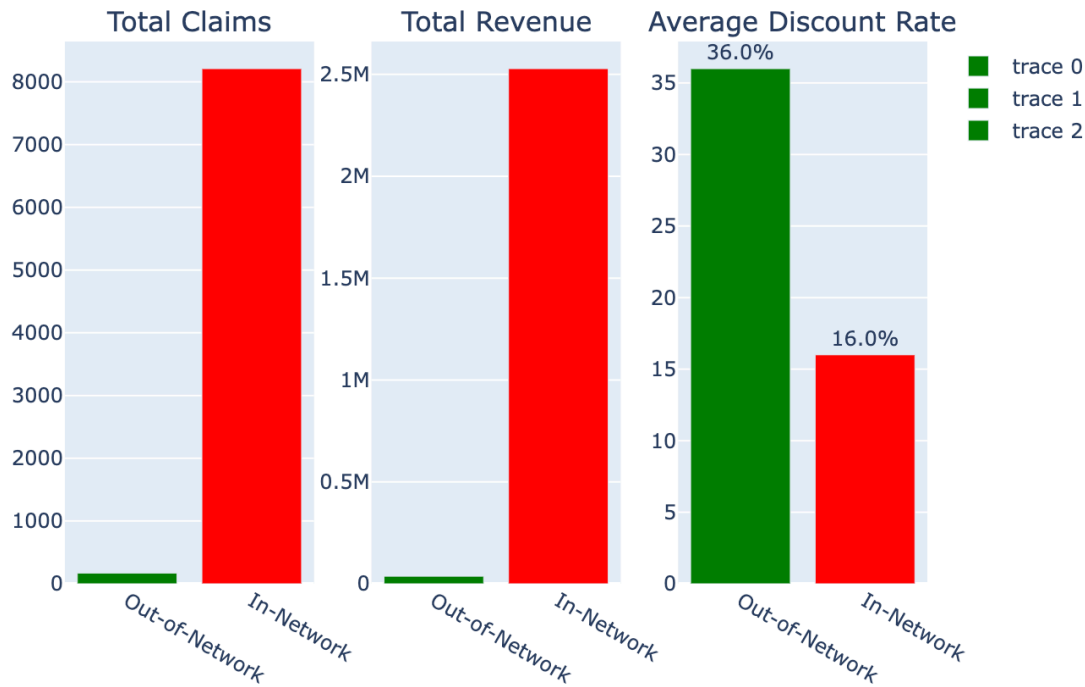
Average Charge by State



Average Charge by Region

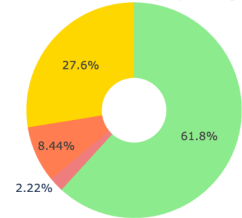


In-Network vs Out-of-Network Analysis

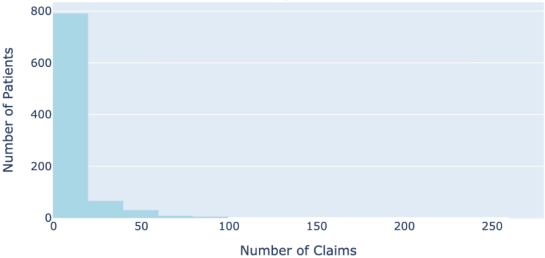


Patient Activity & Segmentation Analysis

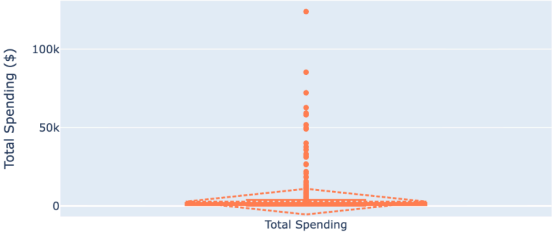
Patient Distribution by Spending Tier



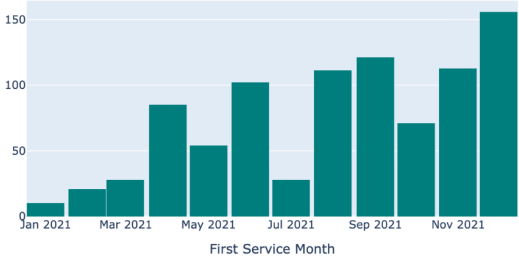
Claims per Patient



Total Patient Spending Distribution



First Service Month Distribution





Healthcare Analytics Executive Report

Generated on: 2026-05-03 01:07:17



Key Performance Indicators

\$2,565,170

Total Gross Revenue

\$1,629,859

Total Net Revenue

1,706

Unique Patients (Multi-Visit)

5,722

COVID-19 Tests Performed

\$1,043,112

COVID-19 Test Revenue

98.0%

In-Network Claims



COVID-19 Testing Summary

claim_code	test_count	total_charge	avg_discount_per_test
87426	86	20053.09	32.52
87635	23	3915.00	23.71
G2023	1389	92203.42	4.66
G2024	27	1381.00	10.29
U0003	1495	403795.46	53.62
U0004	732	356885.25	285.73
U0005	1970	164879.25	4.86

🏆 Top 10 Providers by Revenue

provider_name	total_charged	total_claims	in_network_pct
SIGNAL DIAGNOSTICS LLC	327228.00	2117	100.0
FLOW HEALTH LABORATORIES	327228.00	2117	100.0
BRIO CLINICAL INC	173066.25	174	100.0
ITALIA, HIRENKUMAR MD	170550.00	895	100.0
EHRHART, KEVIN M MD	123940.00	2	100.0
WEST WILSHIRE SURGICAL CENTER	121525.00	18	100.0
WAHLIG, RONALD MD	121525.00	18	100.0
GRIFFITHS, CHESTER F MD	121525.00	18	100.0
BREAKTHROUGHGENOMICS INC	102450.00	538	100.0
ELLENHORN, JOSHUA	62645.80	2	100.0

⚠️ High Discount Alerts (>50%)

procedure_desc	charge_amt	allowed_amt	discount_pct	provider_name
<NA>	45.00	0.0	100.0%	BROFFMAN, JILL MD
<NA>	32.34	0.0	100.0%	ADVANCED HOMECARE
<NA>	300.00	0.0	100.0%	SIGNAL DIAGNOSTICS LLC
<NA>	243.18	0.0	100.0%	ADVANCED HOMECARE
<NA>	300.00	0.0	100.0%	FLOW HEALTH LABORATORIES
<NA>	117.64	0.0	100.0%	ADVANCED HOMECARE
<NA>	75.00	0.0	100.0%	SIGNAL DIAGNOSTICS LLC
<NA>	41.02	0.0	100.0%	ADVANCED HOMECARE
<NA>	75.00	0.0	100.0%	FLOW HEALTH LABORATORIES
<NA>	250.00	0.0	100.0%	GLASER, ELIZABETH B
<NA>	70.38	0.0	100.0%	SIGNAL DIAGNOSTICS LLC
<NA>	250.00	0.0	100.0%	NEUROLOGICAL ASSOCIATES SURGERY CENTER

<NA>	39.75	0.0	100.0%	ADVANCED HOMECARE
<NA>	187.50	0.0	100.0%	NEUROLOGICAL ASSOCIATES SURGERY CENTER
<NA>	117.64	0.0	100.0%	ADVANCED HOMECARE
<NA>	187.50	0.0	100.0%	GLASER, ELIZABETH B
<NA>	243.18	0.0	100.0%	ADVANCED HOMECARE
<NA>	1312.50	0.0	100.0%	NEUROLOGICAL ASSOCIATES SURGERY CENTER

Total alerts: 1851 claims with >50% discount

This report was generated automatically from Amazon Athena healthcare analytics views.

For questions or custom analyses, contact the data engineering team.

 Report saved to s3://hc-pipeline-demo/reports/healthcare_executive_report_20260503_010717.html